

## > Cipher Toolkit

## CAESAR CIPHER

## > Shifting the alphabet

Every letter moves forward through the alphabet by the same fixed amount. Write the alphabet underneath itself, shifted along by your guess, and compare it against the message.

[illegible][illegible][illegible]

## SUBSTITUTION CIPHER

## > Letter-for-letter swaps

Each letter has been swapped for a different letter—not necessarily a fixed shift. Use the frequency table below as a guide, then test your guesses in the workspace.

Letter frequency in English (most common first)

|   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E | T | A | O | I | N | S | H | R | D | L | C | U |
| M | W | F | G | Y | P | B | V | K | J | X | Q | Z |

Workspace: write your guessed mapping underneath each ciphertext letter.

[illegible][illegible][illegible]

## TRANSPOSITION CIPHER

## > Reordering, not replacing

The letters themselves don't change—only their order does. Write the ciphertext into the grid below, one letter per box, left to right, then try reading it back out a different way: down a column, or in a zigzag.

[illegible]

## RAIL FENCE CIPHER

## > A specific zigzag pattern

A message can be written in a zigzag down and up across several rails, then read off rail by rail to produce the ciphertext. To reverse it, place the letters back into the same zigzag shape and read them off in zigzag order.

Worked example (3 rails)—message: MEETMEATMIDNIGHT

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M | . | . | . | M | . | . | . | M | . | . | . | I | . | . | . |
| . | E | . | T | . | E | . | T | . | I | . | N | . | G | . | T |
| . | . | E | . | . | . | A | . | . | . | D | . | . | . | H | . |

Read rail by rail, that becomes: M M M I E T E T I N G T E A D H

Workspace: try laying out a real ciphertext below using the grid from the transposition cipher page, but placing letters along a zigzag instead of straight rows.

[illegible]

## > The mirror alphabet

Every letter swaps with its mirror image in the alphabet—A with Z, B with Y, and so on. No key is needed; just read straight off the strip below.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Z | Y | X | W | V | U | T | S | R | Q | P | O | N | M | L | K | J | I | H | G | F | E | D | C | B | A |

## VIGENÈRE CIPHER

### > Tabula recta

Used when you're given a keyword instead of a fixed shift. Line the keyword up underneath the message, repeating it for the full length. For each letter: find the row labelled with that letter's keyword letter, then look along it to the column labelled with the plaintext letter—the cell where they meet is the ciphertext letter.

Decrypting works the same way in reverse: find the ciphertext letter in that row, then read off its column heading.

|   | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |